

Requirements Documentation	2
Interactive House	2
Revision History	2
Color Code Legend	2
Requirements List	2
Requirements Descriptions	3
R1 – Register user	3
R2 – Authenticate user (login).....	3
R3 – Terminate user session (logout)	3
R4 – Open/close door	3
R5 – Turn light on/off.....	3
R6 – Open/close window	4
R7 – Turn fan on/off	4
R8 – Turn buzzer on/off	4
R9 – Retrieve device status information	4
R10 – Manage authorized devices and units	4
R11 – Reject unauthorized requests.....	4
R12 – Ensure data persistence and recovery.....	4
R13 – Update user password.....	4
R14 – Delete device field	5
R15 – LCD text display	5
Traceability Overview	5

Requirements Documentation

Interactive House

GitHub: [Interactive House](#) — see `SG1_Server_DB/` for server & database code

Revision History

Date	Version	Description	Author
08/02/2026	1.0	Initial functional requirements	Katerina Arvay, Andre Sandblom, Dario Ostojic
01/03/2026	1.1	Requirement updates with new devices	Katerina Arvay, Andre Sandblom, Dario Ostojic
17/03/2026	1.2	Refined functional requirements to align with current system architecture and deployment setup	Katerina Arvay, Andre Sandblom, Dario Ostojic
23/03/2026	1.3	Addition of a backup database	Katerina Arvay, Andre Sandblom, Dario Ostojic
08/05/2026	1.4	Completion status added; R12 corrected (Supabase not implemented); R10, R12 priority updated to Desirable; R13–R15 added; traceability column added	Katerina Arvay, Andre Sandblom, Dario Ostojic
08/05/2026	1.5	Traceability table updated: IDs replaced with plain-language descriptions	Katerina Arvay, Andre Sandblom, Dario Ostojic

Color Code Legend

Essential / Implemented / Passed	Desirable / Partial	Not Implemented
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Requirements List

Requirement Name	Priority	Status
R1. Register user	Essential	Implemented
R2. Authenticate user (login)	Essential	Implemented
R3. Terminate user session (logout)	Essential	Implemented

Requirement Name	Priority	Status
R4. Open/close door	Essential	Implemented
R5. Turn light on/off	Essential	Implemented
R6. Open/close window	Essential	Implemented
R7. Turn fan on/off	Essential	Implemented
R8. Turn buzzer on/off	Essential	Implemented
R9. Retrieve device status information	Essential	Implemented
R10. Manage authorized devices and units	Essential	Partial
R11. Block unauthorized access	Essential	Implemented
R12. Ensure data persistence and recovery	Desirable	Not Implemented
R13. Update user password	Desirable	Implemented
R14. Delete device field	Desirable	Implemented
R15. LCD text display (ledTextDisplay)	Desirable	Implemented

Requirements Descriptions

R1 – Register user

The server allowed new users to create an account in the system. User information was stored permanently in the databases and each user was given appropriate access rights.

R2 – Authenticate user (login)

The server verified user credentials when a user accessed the system. Only users who were successfully logged in were allowed to control devices or view system information.

R3 – Terminate user session (logout)

The server allowed users to log out of the system. Logging out ends the active session and prevents unauthorized use of the system.

R4 – Open/close door

The server allowed users to open or close the door. The door state was stored based on confirmed execution via the gateway.

R5 – Turn light on/off

After a command was processed, the server updated the stored state of the light in the database based on the confirmed result received from the device through the gateway.

R6 – Open/close window

The server ensured that the window state stored in the system always matches the actual state reported by the device, not only the requested command.

R7 – Turn fan on/off

The server received commands to turn a fan device on or off. In case of multiple fan devices, a specific option was provided to turn each on or off independently. After a command was processed, the server updated the stored state of the specific fan in the database based on confirmed execution via the gateway.

R8 – Turn buzzer on/off

The server allowed users to turn the buzzer device on or off. The buzzer state was stored after the command was processed.

R9 – Retrieve device status information

The server allowed authorized users and units to request the current status of registered devices, such as whether a light was on or off or whether an alarm was active.

R10 – Manage authorized devices and units

The server was designed to keep track of which devices and units are registered and authorized to interact with the system. Only devices and units in this registry are allowed to communicate with the server. Note: this requirement was only partially implemented.

R11 – Reject unauthorized requests

The server rejected any requests coming from users, devices or units that were not authenticated or authorized.

R12 – Ensure data persistence and recovery

The server was intended to ensure that all critical system data (users, device states, and configurations) is persistently stored and recoverable in case of failure. Note: this requirement was only partially implemented. The planned secondary database (Supabase) was not implemented — the supabase/ folder contains only placeholder stub code. Data persistence relied entirely on Firebase Firestore.

R13 – Update user password

The server allowed authenticated users to update their password. The current password was verified before the new password was accepted and hashed (bcrypt) before storage.

Traceability: Interaction sequence diagram (password update flow); Test: verify new password works and old password is rejected.

R14 – Delete device field

The server allowed authenticated users to remove a named device field from their document via DELETE /users/device. The password field was explicitly protected and could not be deleted.

Traceability: API endpoint reference; Test: delete a custom device field and verify removal; verify password field cannot be deleted.

R15 – LCD text display

The server allowed authenticated users to set a text message on the house LCD display via PATCH /users/device with deviceName 'ledTextDisplay'. The SG3 gateway forwarded the text to the Arduino LCD1602 via serial command M[text]] (max 16 chars).

Traceability: Use case diagram (LCD action), Interaction sequence diagram; Test: send text message and verify it appears on screen.

Traceability Overview

The table below maps each requirement to its related design item and test, showing the connections between the Requirements, Design, and Test documents.

Requirement	Design Item	Test
Register user	Use case diagram, Interaction sequence diagram	Register with valid credentials; reject duplicate username; reject missing API key
Authenticate user	Interaction sequence diagram	Login with correct password; reject wrong password
Terminate session	Interaction sequence diagram	Token is discarded on logout
Open/close door	Use case diagram, Interaction sequence diagram, State diagram	Send open/close command; verify state is saved and gateway receives it
Turn light on/off	Use case diagram, Interaction sequence diagram, State diagram	Toggle white light on and off; set orange light brightness value
Open/close window	Use case diagram, Interaction sequence diagram, State diagram	Send open/close command; verify gateway receives correct command
Turn fan on/off	Use case diagram, Interaction sequence diagram, State diagram	Control fan A and fan B independently
Turn buzzer on/off	Use case diagram, Interaction sequence diagram, State diagram	Turn buzzer on and off; verify response
Retrieve device status	Interaction sequence diagram	Returned status matches what was set; password never exposed in response
Manage authorized devices	Entity and class diagram, API endpoint reference	Partial — no formal device registry implemented

Requirement	Design Item	Test
Block unauthorized access	Interaction sequence diagram	Reject command without token; reject expired token; verify no state changes occur
Data persistence and recovery	—	Verify states persist after restart — future work, Supabase not implemented
Update user password	Interaction sequence diagram, API endpoint reference	Change password; verify new password works; verify old password is rejected
Delete device field	API endpoint reference	Delete a custom device field and verify removal; verify password field cannot be deleted
LCD text display	Use case diagram, Interaction sequence diagram	Send text message; verify it appears on screen (max 16 characters)